

A Roe-Algebraic Reduction for Doubled-and-Jammed Universal Chern Hamiltonians

Yitzchak Shmalo*

Einstein Institute of Mathematics
Hebrew University of Jerusalem

April 27, 2026

Abstract

This paper isolates the coarse-geometric and K -theoretic part of the doubled-and-jammed construction for universal Chern Hamiltonians on triangulated surfaces with boundary. The finite doubling, jamming, Schur-complement estimates, leakage bounds, and numerical diagnostics are established in the finite-dimensional paper. Here they are used only as input. We form the coarse disjoint union of the closed doubles and the interface ideal $C_u^*(Y \subset X)$ associated to the glued-boundary collar. Whenever the direct-sum closed doubled Hamiltonian is a gapped Roe-algebra element, its bulk Fermi projection defines a class in $K_0(C_u^*(X))$, and the doubled-and-jammed family defines a sidewise class-A interface. The main result identifies the interface class with the connecting image

$$\partial_Y(E_+(\pi(P_\pm))) \in K_1(C_u^*(Y \subset X)).$$

The construction is intrinsic to the quotient extension by the interface ideal. Under a compatible coarse bulk-interface pairing convention and the positive-side Chern compatibility assumption, the paired winding number is ± 1 . Thus the paper proves a conditional Roe-algebraic reduction and requires no separate boundary spectral gap or componentwise edge-mode hypothesis.

1 Introduction

Higson and Prodan construct universal Chern Hamiltonians on closed oriented triangulated surfaces from the simplicial boundary operator and a controlled Poincaré-duality/cap-product operator [2, 3]. The finite-dimensional paper [9] applies that closed model to a triangulated surface with boundary by passing to the closed double and adding a large confining potential on the reflected copy. It proves the finite-dimensional part of the construction: the double is a genuine closed abstract triangulation, the jammed operator admits explicit $1/M$ and $1/M^2$ decoupling estimates, and the original-copy coupling is confined to a fixed collar of the glued boundary.

The present paper does not repeat those finite-dimensional arguments. It starts from their output and proves the coarse interface theorem. The point is to separate two tasks that are logically different:

- (i) finite-dimensional doubling and jamming, which is local linear algebra and is handled in [9];

*Corresponding author: yitzchak.shmalo@gmail.com.

- (ii) the refinement-limit interface class, which is naturally formulated in the Roe algebra of a coarse disjoint union and is handled here.

At the operator-algebra level, the minimal input is that the direct sum of the closed doubled Hamiltonians is a self-adjoint Roe-algebra element with a spectral gap at zero. This is not an avoidable technicality; it is exactly what makes the Fermi projection a continuous functional calculus element of $C_u^*(X)$. In the final theorem this input is not left as an independent boundary hypothesis: it is supplied by the cited Higson–Prodan closed-bulk theorem whenever the closed doubled family satisfies that theorem’s closed-refinement hypotheses.

The second external input is the standard bulk–interface pairing theorem. We construct the interface K_1 -class directly from

$$0 \longrightarrow C_u^*(Y \subset X) \longrightarrow C_u^*(X) \xrightarrow{\pi} C_u^*(X)/C_u^*(Y \subset X) \longrightarrow 0.$$

The assertion that its winding number equals the positive-side Chern number is then the cited coarse bulk–interface theorem, after fixing the algebra comparison and sidewise pairing convention in the Ewert–Meyer Roe-algebra framework [1], with Kubota’s controlled framework providing a parallel Mayer–Vietoris bulk–edge formulation [7]. The paper therefore proves a reduction theorem: closed-bulk Higson–Prodan + doubled-and-jammed Roe reduction + positive-side Chern compatibility + compatible coarse bulk–interface correspondence gives the boundary winding number.

Main theorem, in compressed form. The precise theorem proved below is the following. Assume the finite-dimensional package of Theorem 2.2, the closed-doubled Higson–Prodan closed-refinement hypotheses of Theorem 4.9, a compatible sidewise pairing convention in the sense of Definition 10.1, and the positive-side Chern compatibility Assumption 10.3. Then, for every $M > B_{\text{copy}, -}$,

$$\text{Ind}_{\text{int}}(J_{\pm, M}^\infty; T_\pm^+, T_{\pm, M}^-) = \partial_Y(E_+(\pi(P_\pm))),$$

and the paired winding number of this class is ± 1 . The equality of K -classes is proved internally in Theorem 8.3; the numerical winding statement is the external pairing consequence isolated in Section 10.

2 Imported finite-dimensional input

We recall only the notation needed below. Let $(\mathcal{K}_n)$ be a refinement sequence of compact oriented triangulated surfaces with boundary. The finite-dimensional construction gives the closed double $D(\mathcal{K}_n)$, the Hilbert space

$$\mathcal{H}_n := \ell^2(X_n), \quad X_n := X(D(\mathcal{K}_n)),$$

where X_n is the set of barycenters of all simplices in $D(\mathcal{K}_n)$, and the closed Higson–Prodan Hamiltonians

$$H_{\pm, n} := (B_n + B_n^\dagger) \pm S_n.$$

The barycentric graph metric is used throughout: two simplex barycenters are adjacent precisely when the corresponding simplices are incident by inclusion.

Definition 2.1 (Propagation). A bounded operator T on $\ell^2(X_n)$ has propagation at most R if

$$\langle x, Ty \rangle = 0 \quad \text{whenever} \quad d_{X_n}(x, y) > R.$$

Let $X_{n,\text{orig}}$ denote the original side together with the glued-boundary simplices, and let $X_{n,\text{copy}}$ denote the reflected-copy simplices not contained in the glued boundary subcomplex. Thus

$$\mathcal{H}_n = \mathcal{H}_{n,\text{orig}} \oplus \mathcal{H}_{n,\text{copy}}.$$

With respect to this decomposition,

$$H_{\pm,n} = \begin{pmatrix} A_n & C_n \\ C_n^\dagger & D_n \end{pmatrix}, \quad J_{\pm,n,M} := H_{\pm,n} + MP_{n,\text{copy}}.$$

Theorem 2.2 (Finite-dimensional doubled-and-jammed package, from [9]). *For each n , the following hold.*

- (a) *$D(\mathcal{K}_n)$ is a closed oriented triangulated surface, and the block decomposition above is well-defined.*
- (b) *For every fixed non-real z , the original-side compressed resolvent of $J_{\pm,n,M}$ converges to $(A_n - z)^{-1}$ at rate $O(M^{-1})$, uniformly with explicit constants whenever $\text{dist}(z, \text{spec}(D_n + M)) > 0$.*
- (c) *The first effective original-side operator is*

$$A_{n,M}^{(1)} = A_n - \frac{1}{M} C_n C_n^\dagger,$$

and the corresponding compressed resolvent error improves to $O(M^{-2})$ under the stated large- M separation hypotheses.

- (d) *Bounded-energy eigenvectors have copy-side leakage bounded by*

$$\|\psi_{\text{copy}}\| \leq \frac{\|C_n\|}{\text{dist}([-E_{\max}, E_{\max}], \text{spec}(D_n + M))} \|\psi_{\text{orig}}\|.$$

- (e) *If Γ_n is the glued-boundary barycenter set, then the original-copy block C_n has matrix support in a uniformly bounded neighborhood of Γ_n .*

Remark 2.3. Theorem 2.2 is the only finite-dimensional input used here. The present paper does not use the numerical experiments or the detailed Schur-complement proofs from [9]. For logical independence, the theorem may also be read as an explicit finite-dimensional input hypothesis; a self-contained version can include these estimates as an appendix.

3 Coarse disjoint unions and the interface ideal

Definition 3.1 (Uniformly regular refinement family). The refinement sequence $(\mathcal{K}_n)$ is uniformly regular if the metric spaces X_n have uniformly bounded geometry: for every $r > 0$ there is $N(r) < \infty$ such that

$$|B_{X_n}(x, r)| \leq N(r) \quad \text{for all } n \text{ and all } x \in X_n.$$

Let

$$X := \bigsqcup_{n=1}^{\infty} X_n$$

be a coarse disjoint union, with the original metrics on components and intercomponent distances tending to infinity. Equivalently, for each propagation bound r , only finitely many intercomponent

pairs can lie within distance r . Such finite-initial-component effects are absorbed by the interface ideal in Proposition 3.7. Since the boundary families considered here have nonempty glued boundary, $Y \cap X_n \neq \emptyset$ for each component. Let $\Gamma_n \subset X_n$ be the glued-boundary barycenter set and set

$$\Gamma := \bigsqcup_n \Gamma_n, \quad Y^{(R)} := \bigsqcup_n N_R(\Gamma_n).$$

The scalar/uniform Roe algebra $C_u^*(X)$ is the norm closure of bounded finite-propagation operators on $\ell^2(X)$. Since X is discrete, all bounded finite-propagation operators are locally compact in the scalar representation. In particular, $C_u^*(X)$ is unital.

Remark 3.2 (Roe-algebra convention and comparison with the literature). All Roe algebras in this paper are scalar/uniform, non-equivariant Roe algebras represented on $\ell^2(X)$. This is the convention in which the identity and the side projections are finite-propagation locally compact operators. Some bulk–edge references use stabilized Roe algebras, localization algebras, or equivariant/twisted variants. The final pairing statement below is therefore formulated as an explicit compatibility hypothesis with the chosen coarse bulk–interface theorem, rather than as an automatic identification of all models.

Definition 3.3 (Interface ideal). For $Y \subset X$, the interface ideal $C_u^*(Y \subset X)$ is the norm closure of bounded finite-propagation operators whose matrix support is contained in

$$N_R(Y) \times N_R(Y)$$

for some finite R .

Lemma 3.4 (Ideal property). $C_u^*(Y \subset X)$ is a closed two-sided ideal in $C_u^*(X)$.

Proof. At the algebraic level, support near Y is preserved under adjoints and finite sums. If T has propagation r and S is supported in $N_R(Y) \times N_R(Y)$, then TS and ST are supported in $N_{R+r}(Y) \times N_{R+r}(Y)$. Taking norm closures gives a closed two-sided ideal. $\square$

Proposition 3.5 (Independence of fixed collar radius). For fixed $R, R' \geq 0$,

$$C_u^*(Y^{(R)} \subset X) = C_u^*(Y^{(R')} \subset X).$$

Proof. If $R' \geq R$, then

$$Y^{(R)} \subset Y^{(R')} \subset N_{R'}(Y^{(R)}).$$

Thus the algebraic operators supported near $Y^{(R)}$ are exactly the algebraic operators supported near $Y^{(R')}$. Taking norm closures gives the claim. The case $R > R'$ is symmetric. $\square$

Henceforth we fix one collar $Y = Y^{(R)}$, write

$$\mathcal{A} := C_u^*(X), \quad \mathcal{I} := C_u^*(Y \subset X),$$

and denote the quotient map by $\pi : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{I}$.

Proposition 3.6 (Finite-Hausdorff invariance of the interface ideal). Let $Y_0, Y_1 \subset X$ be subsets at finite Hausdorff distance: for some $R < \infty$,

$$Y_0 \subset N_R(Y_1), \quad Y_1 \subset N_R(Y_0).$$

Then

$$C_u^*(Y_0 \subset X) = C_u^*(Y_1 \subset X).$$

Proof. If an algebraic Roe operator is supported in $N_s(Y_0) \times N_s(Y_0)$, then it is supported in $N_{s+R}(Y_1) \times N_{s+R}(Y_1)$. Hence the algebraic ideal for Y_0 is contained in the algebraic ideal for Y_1 . The reverse inclusion is identical. Taking norm closures proves the claim. $\square$

Proposition 3.7 (Finite initial components). *Assume each $Y \cap X_n$ is nonempty. Any bounded finite-propagation operator supported on only finitely many components X_n belongs to $\mathcal{I}$. Consequently finite-component changes are invisible in the quotient $\mathcal{A}/\mathcal{I}$.*

Proof. If V is supported on $F = \bigsqcup_{n=1}^N X_n$, then F is finite. Since each relevant interface component is nonempty,

$$R_F := \max_{x \in F} \text{dist}(x, Y) < \infty.$$

Thus $F \subset N_{R_F}(Y)$, so V is supported near Y and belongs to $\mathcal{I}$. $\square$

4 The bulk Fermi projection

Assumption 4.1 (Uniform cap-product locality). There are constants $R_S, N_S, c_S < \infty$, independent of n , such that

$$S_n(x, y) = 0 \quad \text{if } d_{X_n}(x, y) > R_S,$$

each row and column of S_n has at most N_S nonzero entries, and

$$|S_n(x, y)| \leq c_S \quad \text{for all } x, y \in X_n.$$

Lemma 4.2 (Uniform norm bound for S_n). *Under Assumption 4.1,*

$$\text{prop}(S_n) \leq R_S, \quad \|S_n\| \leq N_S c_S.$$

Proof. The propagation bound is immediate. For the norm bound, the row and column sums of absolute values are bounded by $N_S c_S$. The Schur test gives $\|S_n\| \leq N_S c_S$. $\square$

Remark 4.3. In applications this assumption is the local-control part of the imported Higson–Prodan cap-product construction on the closed doubles. It is stated separately because the Roe-algebra arguments only use the uniform propagation, row/column support, and coefficient bounds, not the full construction of the Poincaré-duality operator.

Assumption 4.4 (Gapped bulk Roe operator). The direct-sum closed doubled Hamiltonian

$$H_\pm^\infty := \bigoplus_n H_{\pm, n}$$

is a self-adjoint element of $\mathcal{A} = C_u^*(X)$, and there are constants

$$B_{\text{bulk}} > g_{\text{bulk}} > 0$$

such that

$$\text{spec}(H_\pm^\infty) \subset [-B_{\text{bulk}}, -g_{\text{bulk}}] \cup [g_{\text{bulk}}, B_{\text{bulk}}].$$

Proposition 4.5 (Concrete sufficient conditions for the bulk Roe hypothesis). *Assume the family is uniformly regular, Assumption 4.1 holds, and there are constants $B_{\text{bulk}} > g_{\text{bulk}} > 0$ such that for every n ,*

$$\text{spec}(H_{\pm, n}) \subset [-B_{\text{bulk}}, -g_{\text{bulk}}] \cup [g_{\text{bulk}}, B_{\text{bulk}}].$$

Then Assumption 4.4 holds.

Proof. Uniform regularity gives a uniform degree bound for the barycentric graphs, hence a uniform norm and propagation bound for $B_n + B_n^\dagger$. Lemma 4.2 gives a uniform norm and propagation bound for S_n . Thus $H_\pm^\infty = \bigoplus_n H_{\pm,n}$ is a bounded finite-propagation operator on X , so $H_\pm^\infty \in \mathcal{A}$. The displayed componentwise spectral inclusion implies the same inclusion for the direct sum. $\square$

Proposition 4.6 (Roe-algebra bulk Fermi projection). *Assume Assumption 4.4. Then*

$$P_\pm := \chi_{(-\infty, 0)}(H_\pm^\infty)$$

belongs to $\mathcal{A}$. Hence $P_\pm$ defines a class $[P_\pm] \in K_0(\mathcal{A})$.

Proof. Choose $f \in C_c(\mathbb{R})$ such that

$$f(t) = 1 \quad \text{on } [-B_{\text{bulk}}, -g_{\text{bulk}}/2], \quad f(t) = 0 \quad \text{on } [g_{\text{bulk}}/2, B_{\text{bulk}}].$$

Then f agrees with $\chi_{(-\infty, 0)}$ on $\text{spec}(H_\pm^\infty)$. Continuous functional calculus in $\mathcal{A}$ gives

$$P_\pm = f(H_\pm^\infty) \in \mathcal{A}.$$

$\square$

Remark 4.7 (Eventual gaps and finite initial components). If the cited closed-bulk theorem supplies the spectral gap only for all sufficiently large refinement levels, discard the finitely many initial components when forming the direct-sum projection and then reinsert them arbitrarily. The difference between any two choices on finitely many initial components is supported on finitely many components and hence lies in the interface ideal by Proposition 3.7. Therefore the interface class constructed below is unchanged. Thus an eventual closed-bulk gap is enough for the K -theoretic interface class.

Proposition 4.8 (Why a spectral separation is needed). *The spectral-separation input in Assumption 4.4 cannot be deleted in general. There are self-adjoint Roe-algebra elements H with $0 \in \text{spec}(H)$ for which no continuous function f on $\text{spec}(H)$ agrees with $\chi_{(-\infty, 0)}$ on $\text{spec}(H)$.*

Proof. Take $X = \mathbb{N}$ with the usual coarse metric and define a diagonal multiplication operator

$$H e_{2k-1} = -\frac{1}{k} e_{2k-1}, \quad H e_{2k} = \frac{1}{k} e_{2k}.$$

This operator has propagation 0, hence belongs to $C_u^*(X)$. Its spectrum is

$$\{0\} \cup \{\pm 1/k : k \geq 1\}.$$

If f were continuous on $\text{spec}(H)$ and agreed with $\chi_{(-\infty, 0)}$, then

$$f(-1/k) = 1, \quad f(1/k) = 0$$

for all k . Both sequences converge to 0, so continuity at 0 would force $f(0) = 1$ and $f(0) = 0$, a contradiction. Thus the continuous-functional-calculus proof of the Fermi projection requires an actual spectral separation of the negative and nonnegative spectral parts. Assumption 4.4 is a convenient standard form of that separation. $\square$

Theorem 4.9 (Closed-bulk theorem applied to the doubles). *Let $(\mathcal{K}_n)$ be a uniformly regular boundary refinement family satisfying the hypotheses of the finite-dimensional construction. Suppose moreover that the closed doubled refinement family $(D(\mathcal{K}_n))_n$ satisfies the closed-refinement hypotheses of the Higson–Prodan closed-bulk theorem, with the corresponding controlled Poincaré-duality operators and Higson–Prodan Chern pairing. Then Assumption 4.4 holds and*

$$\text{Ch}_{\text{HP}}([P_{\pm}]) = \pm 1.$$

Proof. This is precisely the imported closed-surface result of Higson–Prodan and the underlying Higson–Roe controlled Poincaré-duality machinery [2, 3, 4, 5, 6]: the operators

$$H_{\pm,n} = (B_n + B_n^{\dagger}) \pm S_n$$

are the Higson–Prodan universal Chern Hamiltonians on the closed triangulations $D(\mathcal{K}_n)$. The uniform gap, controlled locality, Fermi projection, and Chern computation are exactly the closed-bulk conclusions of the cited Higson–Prodan theorem under the stated closed-refinement hypotheses. $\square$

Remark 4.10 (Imported Higson–Prodan input). The closed-bulk theorem is used here only through the following data: closed oriented triangulated surfaces $D(\mathcal{K}_n)$; uniform bounded geometry of their barycentric spaces; controlled Poincaré-duality/cap-product operators with uniform propagation and coefficient bounds; refinement admissibility and orientation compatibility in the sense required by Higson–Prodan; and the closed-bulk conclusions consisting of a uniform Fermi gap, the Roe-algebra Fermi projection, and the Higson–Prodan Chern value $\text{Ch}_{\text{HP}}([P_{\pm}]) = \pm 1$. No closed-bulk Chern computation is reproved in this paper.

Proposition 4.11 (A sufficient admissibility-transfer criterion). *Assume the finite-dimensional construction applies to $(\mathcal{K}_n)$, and assume the following properties of the doubles:*

- (a) *$D(\mathcal{K}_n)$ is a closed oriented triangulated surface for every n ;*
- (b) *the barycentric spaces $X(D(\mathcal{K}_n))$ have uniformly bounded geometry;*
- (c) *the controlled Poincaré-duality/cap-product operators used by Higson–Prodan have uniform propagation and coefficient bounds on $D(\mathcal{K}_n)$;*
- (d) *the closed doubled family satisfies the remaining closed-refinement hypotheses required by the cited Higson–Prodan theorem.*

Then Theorem 4.9 applies to $(D(\mathcal{K}_n))_n$.

Proof. Part (a) is the geometric input needed to apply the closed-surface construction. Parts (b) and (c) give the uniform geometric control required for the direct-sum Roe-algebra element. Part (d) is precisely the remaining closed-refinement hypothesis of the cited theorem. Therefore the closed Higson–Prodan theorem applies to the family of doubles, giving Theorem 4.9. $\square$

5 Quotient splitting away from the interface

Let X^+ be the strict original side outside Y , let X^- be the strict reflected-copy side outside Y , and let Q_+, Q_-, Q_Y be the corresponding multiplication projections. After enlarging Y by a fixed collar, which does not change $\mathcal{I}$ by Proposition 3.5, we may assume

$$X = X^+ \sqcup X^- \sqcup Y.$$

Lemma 5.1 (Cross terms are interface-supported). *If T is a bounded finite-propagation operator on $\ell^2(X)$, then*

$$Q_+TQ_-, \quad Q_-TQ_+, \quad Q_YT, \quad TQ_Y \in \mathcal{I}.$$

Proof. The terms involving Q_Y are supported near Y by definition. For Q_+TQ_- , let $\text{prop}(T) \leq r$. Because the intercomponent distances in the coarse disjoint union tend to infinity, all propagation- r matrix coefficients connecting different components are supported on only finitely many components. By Proposition 3.7, those finite-component terms belong to $\mathcal{I}$. After discarding them modulo $\mathcal{I}$, a nonzero matrix coefficient connects points in the same doubled component: one on the strict original side and one on the strict copy side. Any barycentric graph path between such points must pass through the glued-boundary collar. Hence both endpoints lie in a bounded neighborhood of Y , with bound depending only on r . Thus $Q_+TQ_- \in \mathcal{I}$. The proof for Q_-TQ_+ is identical. $\square$

Proposition 5.2 (Central quotient side projections). *Let*

$$e_+ := \pi(Q_+), \quad e_- := \pi(Q_-).$$

Then e_+ and e_- are orthogonal central projections in $\mathcal{A}/\mathcal{I}$, and

$$e_+ + e_- = 1_{\mathcal{A}/\mathcal{I}}.$$

Consequently

$$\mathcal{A}/\mathcal{I} = e_+(\mathcal{A}/\mathcal{I}) \oplus e_-(\mathcal{A}/\mathcal{I}),$$

with central projection maps

$$E_+(b) := e_+b, \quad E_-(b) := e_-b.$$

Proof. For algebraic Roe operators T , Lemma 5.1 gives

$$Q_+T - TQ_+ = Q_+T(Q_- + Q_Y) - (Q_- + Q_Y)TQ_+ \in \mathcal{I}.$$

By norm closure this holds for every $T \in \mathcal{A}$. Therefore e_+ is central in $\mathcal{A}/\mathcal{I}$. The same argument gives centrality of e_- . Orthogonality follows from $Q_+Q_- = 0$, and

$$e_+ + e_- = \pi(Q_+ + Q_-) = \pi(1 - Q_Y) = 1_{\mathcal{A}/\mathcal{I}}$$

because $Q_Y \in \mathcal{I}$. $\square$

Definition 5.3 (Coarsely excisive side decomposition). *The decomposition $X = X^+ \cup X^-$ relative to Y is coarsely excisive if for every $r > 0$ there is $s > 0$ such that*

$$N_r(X^+) \cap N_r(X^-) \subset N_s(Y).$$

Proposition 5.4 (The doubled side decomposition is coarsely excisive). *The strict original/copy decomposition associated to the glued boundary is coarsely excisive.*

Proof. Let $x \in N_r(X^+) \cap N_r(X^-)$. Then there are $x_+ \in X^+$ and $x_- \in X^-$ with $d(x, x_{\pm}) \leq r$. Hence $d(x_+, x_-) \leq 2r$. Any barycentric graph path from the strict original side to the strict copy side must pass through a simplex meeting the glued boundary, hence through $\Gamma \subset Y$. Therefore $x_+ \in N_{2r}(Y)$, and so

$$x \in N_{3r}(Y).$$

Thus one may take $s = 3r$. $\square$

6 Abstract quotient Fermi calculus

The next facts are purely C^* -algebraic. They make explicit that the interface class only needs a quotient gap and a central side decomposition; gapped comparison lifts are a convenient way to compute the quotient projection, not part of the abstract mechanism.

Definition 6.1 (Quotient-gapped self-adjoint element). Let B be a unital C^* -algebra and let $b = b^* \in B$. We say that b is *gapped at zero* if $0 \notin \text{spec}_B(b)$. Its negative spectral projection is

$$p_-(b) := \chi_{(-\infty, 0)}(b) \in B,$$

defined by continuous functional calculus on $\text{spec}_B(b)$.

Lemma 6.2 (Functional calculus in a central direct sum). *Let B be a unital C^* -algebra and let $e_+, e_- \in B$ be orthogonal central projections with $e_+ + e_- = 1$. If $b = b^* \in B$, then*

$$b = e_+ b + e_- b$$

under the C^ -direct sum decomposition $B = e_+ B \oplus e_- B$. If both corner elements $e_{\pm} b$ are gapped at zero in their unital corner algebras $e_{\pm} B$, then b is gapped at zero in B , and*

$$p_-(b) = p_-(e_+ b) + p_-(e_- b).$$

Proof. Centrality and $e_+ + e_- = 1$ give the direct-sum decomposition. Under the canonical isomorphism

$$B \cong e_+ B \oplus e_- B, \quad b \mapsto (e_+ b, e_- b),$$

the spectrum is the union of the two corner spectra and continuous functional calculus is computed componentwise. Therefore the negative spectral projection of b is the direct sum of the two corner negative spectral projections. $\square$

Proposition 6.3 (Quotient-only interface class). *Let*

$$0 \rightarrow I \rightarrow A \xrightarrow{\pi} B \rightarrow 0$$

be a unital extension, and suppose B has central side projections e_+, e_- with $e_+ + e_- = 1$. If $j = j^ \in B$ has both corners $e_{\pm} j$ gapped at zero, then*

$$p_j := p_-(e_+ j) + p_-(e_- j)$$

is a projection in B , and

$$\partial[p_j] \in K_1(I)$$

is a well-defined quotient interface class. No gapped lift of j is required.

Proof. By Lemma 6.2, $p_j = p_-(j)$ is a projection in B . The six-term exact sequence of the extension supplies the connecting map $\partial : K_0(B) \rightarrow K_1(I)$, so $\partial[p_j]$ is defined. The construction uses only the quotient element j and the central side decomposition. $\square$

Proposition 6.4 (Comparison lifts recover the quotient-only class). *In the setting of Proposition 6.3, assume there are self-adjoint lifts $T^{\pm} \in A$ with $0 \notin \text{spec}_A(T^{\pm})$ and*

$$e_+ j = e_+ \pi(T^+), \quad e_- j = e_- \pi(T^-).$$

Let $P^{\pm} = \chi_{(-\infty, 0)}(T^{\pm})$. Then $e_{\pm} j$ are gapped at zero in their corners and

$$p_j = e_+ \pi(P^+) + e_- \pi(P^-).$$

Proof. Let $h_{\pm}$ be continuous functions agreeing with $\chi_{(-\infty,0)}$ on $\text{spec}_A(T^{\pm})$. Since $\text{spec}(\pi(T^{\pm})) \subset \text{spec}_A(T^{\pm})$,

$$\pi(P^{\pm}) = h_{\pm}(\pi(T^{\pm})).$$

Functional calculus in the central corner $e_{\pm}B$ gives

$$p_{-}(e_{\pm}\pi(T^{\pm})) = e_{\pm}h_{\pm}(\pi(T^{\pm})) = e_{\pm}\pi(P^{\pm}).$$

The displayed sidewise identities then give the formula for p_j . □

7 Sidewise interfaces and the boundary class

Definition 7.1 (Sidewise class-A interface). Let $J, T^{+}, T^{-} \in \mathcal{A}$ be bounded self-adjoint operators. We say that $(J; T^{+}, T^{-})$ is a sidewise class-A interface along Y if:

- (a) $0 \notin \text{spec}(T^{\pm})$;
- (b) in the quotient,

$$E_{+}(\pi(J)) = E_{+}(\pi(T^{+})), \quad E_{-}(\pi(J)) = E_{-}(\pi(T^{-})).$$

Proposition 7.2 (Quotient Fermi projection of a sidewise interface). *Let $(J; T^{+}, T^{-})$ be a sidewise class-A interface. Put*

$$P^{\pm} := \chi_{(-\infty,0)}(T^{\pm}).$$

Then $P^{\pm} \in \mathcal{A}$, the quotient element $\pi(J)$ is gapped at zero, and

$$p_J := E_{+}(\pi(P^{+})) + E_{-}(\pi(P^{-}))$$

is the negative spectral projection $p_{-}(\pi(J))$ in $\mathcal{A}/\mathcal{I}$. In particular it depends only on the quotient sidewise data.

Proof. Since $T^{\pm} \in \mathcal{A}$ are self-adjoint and gapped at zero, choose continuous compactly supported cutoffs $h_{\pm}$ agreeing with $\chi_{(-\infty,0)}$ on $\text{spec}(T^{\pm})$. Then

$$P^{\pm} = h_{\pm}(T^{\pm}) \in \mathcal{A}.$$

Apply Proposition 6.4 to the extension

$$0 \rightarrow \mathcal{I} \rightarrow \mathcal{A} \rightarrow \mathcal{A}/\mathcal{I} \rightarrow 0$$

with $B = \mathcal{A}/\mathcal{I}$, $j = \pi(J)$, and $e_{\pm} = \pi(Q_{\pm})$. The sidewise quotient identities are exactly the hypotheses of that proposition. Hence $\pi(J)$ is quotient-gapped and

$$p_{-}(\pi(J)) = E_{+}(\pi(P^{+})) + E_{-}(\pi(P^{-})).$$

□

Definition 7.3 (Interface class). For a sidewise class-A interface, define

$$\text{Ind}_{\text{int}}(J; T^{+}, T^{-}) := \partial_Y([p_J]) \in K_1(\mathcal{I}),$$

where

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \longrightarrow K_1(\mathcal{I})$$

is the connecting map for

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{A} \longrightarrow \mathcal{A}/\mathcal{I} \longrightarrow 0.$$

Remark 7.4. We use the convention that if $q \in \mathcal{A}/\mathcal{I}$ is represented by a self-adjoint lift $\tilde{q} \in \mathcal{A}$, then

$$\partial_Y([q]) = [\exp(2\pi i \tilde{q})] \in K_1(\mathcal{I}),$$

because $\exp(2\pi i \tilde{q}) - 1 \in \mathcal{I}$. Some references use the opposite sign.

Proposition 7.5 (Lifted exponential formula). *Let $q \in \mathcal{A}/\mathcal{I}$ be a projection and let $\tilde{q} = \tilde{q}^* \in \mathcal{A}$ be any self-adjoint lift of q . Then*

$$u_{\tilde{q}} := \exp(2\pi i \tilde{q})$$

belongs to $\mathbf{1} + \mathcal{I}$, and its class in $K_1(\mathcal{I})$ is the connecting image $\partial_Y([q])$. The class is independent of the chosen self-adjoint lift.

Proof. Since $\pi(\tilde{q}) = q$ and $q^2 = q$,

$$\pi(u_{\tilde{q}}) = \exp(2\pi i q) = \mathbf{1}_{\mathcal{A}/\mathcal{I}}.$$

Therefore $u_{\tilde{q}} - \mathbf{1} \in \ker \pi = \mathcal{I}$, so $u_{\tilde{q}} \in \mathbf{1} + \mathcal{I}$. This is the standard exponential representative of the boundary map for a projection in a quotient algebra.

If $\tilde{q}_0$ and $\tilde{q}_1$ are two self-adjoint lifts, then $\tilde{q}_1 - \tilde{q}_0 \in \mathcal{I}$, and

$$\tilde{q}_t := (1 - t)\tilde{q}_0 + t\tilde{q}_1, \quad 0 \leq t \leq 1,$$

is a norm-continuous path of self-adjoint lifts of q . Hence $\exp(2\pi i \tilde{q}_t)$ is a norm-continuous path in $\mathbf{1} + \mathcal{I}$ joining the two unitaries. Their $K_1(\mathcal{I})$ -classes agree. $\square$

8 The doubled-and-jammed interface theorem

Let

$$Q_{\text{copy}} := \bigoplus_n P_{n,\text{copy}}$$

be the strict reflected-copy projection. For scalar jamming $M > 0$, set

$$J_{\pm,M}^{\infty} := H_{\pm}^{\infty} + M Q_{\text{copy}}.$$

Let $H_{\pm,\text{copy}}^{\infty}$ be the copy-side compression of $H_{\pm}^{\infty}$:

$$H_{\pm,\text{copy}}^{\infty} := Q_{\text{copy}} H_{\pm}^{\infty} Q_{\text{copy}} \quad \text{on } Q_{\text{copy}} \ell^2(X).$$

Define the negative-side comparison operator

$$T_{\pm,M}^{-} := Q_{\text{orig}} + H_{\pm,\text{copy}}^{\infty} + M Q_{\text{copy}},$$

where $Q_{\text{orig}} := 1 - Q_{\text{copy}}$. The positive-side comparison operator is

$$T_{\pm}^{+} := H_{\pm}^{\infty}.$$

Finally set

$$B_{\text{copy},-} := \sup_n \max\{0, -\inf \text{spec}(P_{n,\text{copy}} H_{\pm,n} P_{n,\text{copy}} : \mathcal{H}_{n,\text{copy}} \rightarrow \mathcal{H}_{n,\text{copy}})\}.$$

This number is finite under Assumption 4.4.

Remark 8.1. The projection Q_{copy} is the copy-sector projection from the finite-dimensional block decomposition, whereas Q_- is the strict negative-side projection after the interface collar has been absorbed into Y . Their difference is supported in the fixed collar, hence lies in the interface ideal after multiplication by bounded finite-propagation operators. All quotient statements below are therefore made through E_- , not by identifying Q_{copy} with Q_- on the nose.

Proposition 8.2 (Positive margin on the jammed side). *If $M > B_{\text{copy},-}$, then*

$$T_{\pm,M}^- \geq \min\{1, M - B_{\text{copy},-}\} \mathbf{1}.$$

In particular,

$$\chi_{(-\infty,0)}(T_{\pm,M}^-) = 0.$$

Proof. On the original sector, $T_{\pm,M}^- = 1$. On the copy sector,

$$T_{\pm,M}^- = H_{\pm,\text{copy}}^\infty + M,$$

whose spectrum is bounded below by $M - B_{\text{copy},-}$. The two sector bounds give the claim. $\square$

Theorem 8.3 (Doubled-and-jammed Roe interface class). *Assume Assumption 4.4. If $M > B_{\text{copy},-}$, then*

$$(J_{\pm,M}^\infty; T_\pm^+, T_{\pm,M}^-)$$

is a sidewise class-A interface along Y , and

$$\text{Ind}_{\text{int}}(J_{\pm,M}^\infty; T_\pm^+, T_{\pm,M}^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(\mathcal{I}).$$

Proof. Theorem 2.2(e) says that the original-copy coupling is supported in a uniformly bounded collar of the glued boundary. Hence every original-copy cross term belongs to $\mathcal{I}$. Moreover Q_{copy} vanishes on the strict original side and Q_{orig} vanishes on the strict copy side, in the quotient sense encoded by $E_\pm$. Therefore

$$E_+(\pi(J_{\pm,M}^\infty)) = E_+(\pi(H_\pm^\infty + MQ_{\text{copy}})) = E_+(\pi(H_\pm^\infty)) = E_+(\pi(T_\pm^+)),$$

and, after deleting the interface-supported cross terms,

$$E_-(\pi(J_{\pm,M}^\infty)) = E_-(\pi(H_{\pm,\text{copy}}^\infty + MQ_{\text{copy}})) = E_-(\pi(T_{\pm,M}^-)).$$

Here the original-sector identity summand in $T_{\pm,M}^-$ has zero E_- -part. These are exactly the quotient identities in Definition 7.1.

The operator $T_\pm^+$ is gapped at zero by Assumption 4.4. The operator $T_{\pm,M}^-$ is strictly positive by Proposition 8.2. Therefore the triple is a sidewise class-A interface.

By Proposition 7.2,

$$pJ_{\pm,M}^\infty = E_+(\pi(P_\pm)) + E_-(\pi(\chi_{(-\infty,0)}(T_{\pm,M}^-))).$$

The second term is zero by Proposition 8.2. Applying the boundary map gives the displayed formula. $\square$

Corollary 8.4 (Independence of the scalar jam scale). *For any $M_0, M_1 > B_{\text{copy},-}$,*

$$\text{Ind}_{\text{int}}(J_{\pm,M_0}^\infty; T_\pm^+, T_{\pm,M_0}^-) = \text{Ind}_{\text{int}}(J_{\pm,M_1}^\infty; T_\pm^+, T_{\pm,M_1}^-).$$

Proof. Both classes equal $\partial_Y(E_+(\pi(P_\pm)))$ by Theorem 8.3. $\square$

Corollary 8.5 (General positive copy-side confinement). *Let $W = W^* \in \mathcal{A}$ be supported on the strict copy sector and satisfy*

$$W \geq M_0 Q_{\text{copy}}, \quad M_0 > B_{\text{copy},-}.$$

Then

$$J_{\pm, W}^\infty := H_\pm^\infty + W$$

with negative-side comparison operator

$$T_{\pm, W}^- := Q_{\text{orig}} + H_{\pm, \text{copy}}^\infty + W$$

defines the same interface class:

$$\text{Ind}_{\text{int}}(J_{\pm, W}^\infty; T_\pm^+, T_{\pm, W}^-) = \partial_Y(E_+(\pi(P_\pm))).$$

Proof. The proof of Theorem 8.3 uses only the quotient sidewise agreement and the strict positivity of the negative-side comparison. The same arguments apply with MQ_{copy} replaced by W , because the copy-side spectrum is shifted up by at least M_0 . $\square$

Corollary 8.6 (Homotopy of positive copy-side confinements). *Let $s \mapsto W_s$, $s \in [0, 1]$, be a norm-continuous path of self-adjoint copy-supported elements of $\mathcal{A}$, and assume*

$$W_s \geq M_0 Q_{\text{copy}}, \quad M_0 > B_{\text{copy},-},$$

for all s . Then the doubled-and-jammed interface class of $H_\pm^\infty + W_s$ is independent of s .

Proof. By Corollary 8.5, every s gives the same class $\partial_Y(E_+(\pi(P_\pm)))$. $\square$

9 Robustness

Proposition 9.1 (Interface-supported perturbations). *Let $(J; T^+, T^-)$ be a sidewise class-A interface along Y . If $V = V^* \in \mathcal{I}$, then $(J + V; T^+, T^-)$ is again a sidewise class-A interface and*

$$\text{Ind}_{\text{int}}(J + V; T^+, T^-) = \text{Ind}_{\text{int}}(J; T^+, T^-).$$

Proof. Since $V \in \mathcal{I}$, $\pi(J + V) = \pi(J)$. The sidewise quotient identities in Definition 7.1 are therefore unchanged. The comparison operators $T^\pm$, and hence their gaps and quotient Fermi projections, are also unchanged. Thus the projection p_J is unchanged, and so is its boundary class. $\square$

Proposition 9.2 (Homotopy invariance). *Let $s \mapsto (J_s; T_s^+, T_s^-)$, $s \in [0, 1]$, be a norm-continuous family of sidewise class-A interfaces along Y . Assume the comparison gaps are uniform:*

$$\text{dist}(0, \text{spec}(T_s^\pm)) \geq \gamma > 0 \quad \text{for all } s \in [0, 1].$$

Then $\text{Ind}_{\text{int}}(J_s; T_s^+, T_s^-)$ is independent of s .

Proof. Choose a single continuous cutoff h agreeing with $\chi_{(-\infty, 0)}$ on the uniformly gapped spectral windows of all $T_s^\pm$. Then $h(T_s^\pm)$ are norm-continuous projection paths in $\mathcal{A}$, hence

$$p_s = E_+(\pi(h(T_s^+))) + E_-(\pi(h(T_s^-)))$$

is a norm-continuous projection path in $\mathcal{A}/\mathcal{I}$. Its K_0 -class is constant, and applying the connecting map gives the result. $\square$

10 Chern and winding-number interpretation

Definition 10.1 (Sidewise pairing convention). For the final numerical interpretation, we use the Ewert–Meyer Roe-algebraic coarse bulk–interface framework. Thus the scalar/uniform Roe extension used here is identified, after the required Morita/stabilization comparison maps, with the bulk and interface algebras to which the coarse Mayer–Vietoris boundary map applies. Concretely, the convention includes comparison maps making the diagram

$$\begin{array}{ccc} K_0(E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I})) & \xrightarrow{\partial_Y} & K_1(\mathcal{I}) \\ \Phi_{\text{bulk}} \downarrow & & \downarrow \Phi_{\text{edge}} \\ K_0(\mathcal{A}_{\text{bulk}}) & \xrightarrow{\partial_{\text{MV}}} & K_1(\mathcal{A}_{\text{edge}}) \end{array}$$

commute, where the bottom row is the Ewert–Meyer coarse Mayer–Vietoris bulk–interface map. We write

$$\text{Ch}_+ : K_0(E_+(\mathcal{A}/\mathcal{I})) \longrightarrow \mathbb{Z}, \quad \text{Ch}_- : K_0(E_-(\mathcal{A}/\mathcal{I})) \longrightarrow \mathbb{Z},$$

and

$$\text{Wind}_Y : K_1(\mathcal{I}) \longrightarrow \mathbb{Z}$$

for the positive-side Chern pairing, negative-side Chern pairing, and interface winding pairing in that framework, transported through Φ_{bulk} and Φ_{edge} . The sign convention is fixed by

$$\text{Wind}_Y(\partial_Y(p_+)) = \text{Ch}_+(p_+)$$

when the negative-side class is zero.

Theorem 10.2 (Imported Ewert–Meyer sidewise coarse bulk–interface theorem). *In the pairing convention of Definition 10.1, let*

$$p_+ \in K_0(E_+(\mathcal{A}/\mathcal{I})), \quad p_- \in K_0(E_-(\mathcal{A}/\mathcal{I})).$$

If the side decomposition is coarsely excisive, then

$$\text{Wind}_Y(\partial_Y(p_+ + p_-)) = \text{Ch}_+(p_+) - \text{Ch}_-(p_-).$$

In particular, if $p_- = 0$, then

$$\text{Wind}_Y(\partial_Y(p_+)) = \text{Ch}_+(p_+).$$

Proof. This is the Ewert–Meyer coarse bulk–interface correspondence, written in the sidewise quotient notation used here. Ewert–Meyer formulate the Roe-algebra model and identify the coarse Mayer–Vietoris boundary map as the bulk–edge map, including the extension-boundary description used here [1, Propositions 3.2.1–3.2.2 and Corollary 3.2.3]. Kubota’s controlled framework gives a parallel twisted-equivariant formulation of the same Mayer–Vietoris bulk–edge principle [7, Theorem 4.3]. Definition 10.1 is the compatibility datum needed to translate the Ewert–Meyer statement into the present scalar, non-equivariant convention. $\square$

Assumption 10.3 (Positive-side Higson–Prodan Chern compatibility). For the closed doubled family under consideration, the sidewise Chern pairing of Definition 10.1 is compatible with the Higson–Prodan closed-bulk pairing on the positive quotient side:

$$\text{Ch}_+(E_+(\pi(P_{\pm}))) = \pm 1.$$

Remark 10.4. Assumption 10.3 is not a spectral-gap hypothesis and is not an additional boundary Hamiltonian assumption. It is the precise pairing-compatibility statement needed to turn the internally constructed quotient class $E_+(\pi(P_\pm))$ into the numerical Chern value used by the external coarse bulk–interface theorem. In particular, it does not assert $E_-(\pi(P_\pm)) = 0$; the jammed negative comparison is trivial, not the negative quotient of the closed doubled Fermi projection.

Proposition 10.5 (Positive-side Chern bridge). *Assume Definition 10.1, Assumption 10.3, and coarse excision of the side decomposition. Then*

$$\text{Wind}_Y(\partial_Y(E_+(\pi(P_\pm)))) = \pm 1.$$

Proof. Apply Theorem 10.2 with

$$p_+ = E_+(\pi(P_\pm)), \quad p_- = 0.$$

The theorem gives

$$\text{Wind}_Y(\partial_Y(E_+(\pi(P_\pm)))) = \text{Ch}_+(E_+(\pi(P_\pm))),$$

and Assumption 10.3 identifies the right-hand side with ± 1 . $\square$

Theorem 10.6 (Conditional winding theorem under positive-side Chern compatibility). *Let $(\mathcal{K}_n)$ be a uniformly regular boundary refinement family such that the finite-dimensional construction applies, and assume that the closed doubled refinement family $(D(\mathcal{K}_n))_n$ satisfies the cited Higson–Prodan closed-refinement hypotheses. For every scalar $M > B_{\text{copy},-}$, the doubled-and-jammed family has interface class*

$$\text{Ind}_{\text{int}}(J_{\pm,M}^\infty; T_\pm^+, T_{\pm,M}^-) = \partial_Y(E_+(\pi(P_\pm))).$$

If, in addition, Assumption 10.3 holds for the chosen sidewise pairing convention, then this class has winding number ± 1 .

Proof. Theorem 4.9 gives the uniform bulk gap and identifies $P_\pm$ as the Higson–Prodan closed-bulk Fermi projection with $\text{Ch}_{\text{HP}}([P_\pm]) = \pm 1$. The interface class formula is Theorem 8.3. Proposition 5.4 gives coarse excision, and Proposition 10.5 gives

$$\text{Wind}_Y\left(\text{Ind}_{\text{int}}(J_{\pm,M}^\infty; T_\pm^+, T_{\pm,M}^-)\right) = \pm 1$$

under Assumption 10.3. $\square$

Corollary 10.7 (No additional boundary spectral hypothesis). *For every uniformly regular boundary refinement family satisfying the finite-dimensional setup and the closed-doubled Higson–Prodan closed-refinement hypotheses, and for every $M > B_{\text{copy},-}$,*

$$\text{Ind}_{\text{int}}(J_{\pm,M}^\infty; T_\pm^+, T_{\pm,M}^-) = \partial_Y(E_+(\pi(P_\pm))),$$

and, under Assumption 10.3, the standard coarse bulk–interface pairing evaluates this class as ± 1 . No separate uniform boundary spectral gap, interface spectral gap, or componentwise edge-mode hypothesis is required; the numerical winding conclusion is conditional on the sidewise pairing convention and positive-side Chern compatibility.

Proof. This is Theorem 10.6. The closed-bulk gap and Chern number come from the cited Higson–Prodan theorem after doubling; the negative side is made positive by the explicit jamming bound; coarse excision is proved directly in Proposition 5.4; and the numerical winding value is exactly Proposition 10.5. $\square$

Corollary 10.8 (Robustness of the winding value). *Assume the hypotheses of Theorem 10.6, including Assumption 10.3. Then the winding number ± 1 is unchanged under the following operations, whenever the stated sidewise gap hypotheses remain valid:*

- (a) *replacing the scalar jam scale M by any other $M' > B_{\text{copy},-}$;*
- (b) *replacing scalar jamming by any copy-supported confinement W satisfying $W \geq M_0 Q_{\text{copy}}$ and $M_0 > B_{\text{copy},-}$;*
- (c) *adding any self-adjoint interface-supported perturbation $V \in \mathcal{I}$ to the interface operator;*
- (d) *changing the matrices on finitely many initial refinement components;*
- (e) *passing through any norm-continuous sidewise homotopy with a uniform comparison gap.*

Proof. Theorem 10.6 and Proposition 10.5 give

$$\text{Wind}_Y(\partial_Y(E_+(\pi(P_\pm)))) = \pm 1.$$

Corollary 8.4 identifies all sufficiently large scalar jamming classes with this same $K_1(\mathcal{I})$ -class. Corollary 8.5 does the same for every sufficiently positive copy-side confinement. Proposition 9.1 shows that interface-supported perturbations do not change the quotient class, and Proposition 3.7 puts finite initial-component changes inside the interface ideal. Finally, Proposition 9.2 gives invariance under norm-continuous sidewise homotopies with uniform comparison gaps. Since Wind_Y is a homomorphism on $K_1(\mathcal{I})$, all these operations preserve the value ± 1 . $\square$

11 Conclusion

This paper removes the finite-dimensional repetition from the doubled-and-jammed program. The finite-dimensional paper proves the local doubling and jamming estimates. The present paper proves the Roe-algebra interface theorem built from those estimates. The internal reduction is:

$$\text{uniform closed doubled-bulk gap} \Rightarrow [P_\pm] \in K_0(C_u^*(X)) \Rightarrow \partial_Y(E_+(\pi(P_\pm))) \in K_1(C_u^*(Y \subset X)).$$

Combining this reduction with the cited Higson–Prodan closed theorem, the positive-side Chern compatibility statement, and a compatible coarse bulk–interface pairing theorem gives the winding conclusion for boundary refinement families: for all sufficiently large jamming strengths, the interface class has winding number ± 1 in the fixed pairing convention. This winding value is stable under sufficiently positive copy-side confinements, interface-supported perturbations, finite initial-component changes, and norm-continuous sidewise homotopies preserving the comparison gap.

Code and numerical material. The finite-dimensional implementation and numerical scripts for the doubled-and-jammed construction are available in the [Doubling-and-Jamming GitHub repository](#).

Acknowledgments

The research presented in this paper was supported by the European Research Council (ERC) under the European Union’s Horizon 2022 research and innovation program (grant agreement

No. 101041711), by the Simons Foundation (as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning), by the Israel Science Foundation (grant number 2258/19), by the Israel Science Foundation (ISF Grant 4101/25), and by the U.S. National Science Foundation (NSF Grant OISE-2401227).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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